NESTED LOOP CALENDAR GENERATOR

Python Programming — Activity-Based Learning (ABL) Assignment

Student Name:	_____
Roll Number:	_____
Branch / Section:	_____
Course:	Python Programming

1. Aim

To develop a Python program that generates a neatly aligned monthly calendar using the number of days in a month and the weekday on which the month starts. Nested loops and conditional statements are used to format the dates into weeks.

2. Problem Statement

Create a formatted monthly calendar based on user-supplied starting weekday and number of days. The program must align dates under Sunday through Saturday, handle month lengths from 28 to 31 days, and display an error message for invalid inputs.

3. Objectives

• Accept the number of days and starting weekday from the user.

• Validate the day count (28–31) and weekday index (0–6).

• Align the first date beneath the correct weekday.

• Use loops and a conditional statement to print the calendar week by week.

4. Assumptions

Weekday numbering is Sunday = 0, Monday = 1, Tuesday = 2, Wednesday = 3, Thursday = 4, Friday = 5 and Saturday = 6. The input day count is between 28 and 31. This simple generator does not determine the month or year automatically.

5. Algorithm

1. Start the program.

2. Read the number of days and starting weekday.

3. If days are outside 28–31, print an invalid-day message. Otherwise, if weekday is outside 0–6, print an invalid-weekday message.

4. Print the weekday headings.

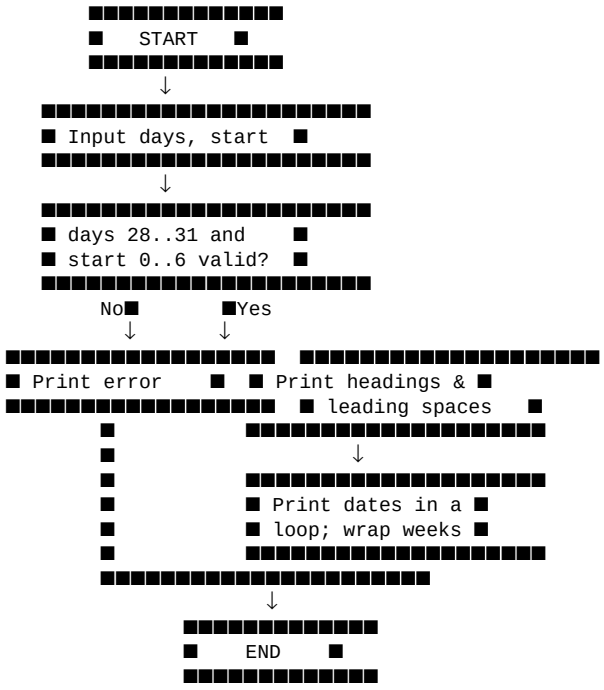
5. Print blank spaces for weekdays before day 1.

6. Loop from day 1 through the final day; print each day in a fixed-width field.

7. After every Saturday, move to the next line.

8. Print a final newline if the last week is incomplete, then stop.

6. Flowchart



7. Python Source Code

```
# Nested Loop Calendar Generator
# Sunday = 0, Monday = 1, ..., Saturday = 6

days = int(input("Enter number of days in the month (28-31): "))
start = int(input("Enter starting weekday (0=Sun, 1=Mon, ..., 6=Sat): "))

if days < 28 or days > 31:
    print("Invalid number of days. Enter a value from 28 to 31.")
elif start < 0 or start > 6:
    print("Invalid starting weekday. Enter a value from 0 to 6.")
else:
    print("\n Sun Mon Tue Wed Thu Fri Sat")
    print("----------------------------")

    # Print blank spaces before the first day
    for i in range(start):
        print("    ", end="")

    # Print each date and start a new line after Saturday
    for day in range(1, days + 1):
        print(f"{day:4}", end="")
        if (start + day) % 7 == 0:
            print()

    # Finish the final week on a new line if necessary
    if (start + days) % 7 != 0:
        print()
```

8. Sample Test Case

Test	Days	Start weekday	Expected result
1	30	4 (Thursday)	Calendar displayed with day 1 under Thursday
2	27	0 (Sunday)	Invalid number of days message
3	30	7	Invalid starting weekday message

9. Sample Input and Output

Input values: 30 days; starting weekday = 4 (Thursday).

10. Conclusion

The Nested Loop Calendar Generator demonstrates how loops, conditional statements and fixed-width formatting can be combined to produce a readable monthly calendar. Input validation helps prevent unsupported day counts and weekday values.

Screenshot note: Run the supplied program in Visual Studio Code and insert an actual terminal screenshot here if your instructor requires proof of execution.